

Simulation Data Representativeness Module (SDRM)

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Architecture Ecosystem: Structured Reality Standards™

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Governed Space: Simulation Data Representativeness

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Canonical Language: English (UCL™)

Governed Space

Simulation Data Representativeness

Canonical Definition

Simulation Data Representativeness Module (SDRM) defines the governance framework through which identified Simulation Inputs and datasets are assessed to determine whether they sufficiently represent the populations, environments, operating states, behaviors, distributions, conditions, variations, and other materially relevant characteristics of the reality that a defined simulation intends to represent.

SDRM establishes the representativeness layer of SIDS by determining whether available information is sufficiently aligned with the Simulation Object, Purpose, Scope, Reality Representation, Assumptions, Model & Environment, and intended simulation context before that information is relied upon as a representative basis for simulation activity.

Operational Role

SDRM serves as the second internal governance space of the Simulation Input & Data Standard (SIDS).

The preceding module, SIIM — Simulation Input Identification Module,

establishes what the material Simulation Inputs are, where they originate, how they are classified, and what simulation roles they may influence.

SDRM asks:

Does the identified information sufficiently represent the reality, population, environment, state, behavior, distribution, or condition that the simulation intends it to represent?

The governed progression is:

Identified Simulation Input → Representation Target → Representativeness
Dimensions → Coverage Assessment → Gap Identification → Materiality Assessment
→ Representativeness Status → Representativeness Conditions & Restrictions

SDRM therefore prevents identified data from being treated as representative merely because it is available, accurate within its original context, technically compatible, or large in volume.

Module Operational Space

SDRM governs: representation target identification, population representativeness, environmental representativeness, geographic representativeness, operational-state representativeness, behavioral representativeness, system-state representativeness, configuration representativeness, distributional representativeness, range representativeness, variation representativeness, rare-condition representation, subgroup representation, class representation, boundary-condition representation, sampling coverage, representation imbalance, representation gaps, overrepresentation, underrepresentation, proxy representation, cross-context applicability, representativeness materiality, representativeness uncertainty, representativeness restrictions, representativeness status, representativeness change, and representativeness traceability.

Module Function

SDRM exists to prevent: available data being assumed representative by default, large datasets being confused with representative datasets, one population being silently generalized to another, normal-operation data being generalized to failure conditions, one geographic environment being generalized globally, one system version being treated as representative of materially different versions, averages concealing relevant distributions, majority populations hiding minority or rare cases, proxy data being treated as direct representation, historical operating conditions being generalized to structurally different environments, missing representation being interpreted as absence of the represented phenomenon, and simulation outputs appearing broadly applicable when the input basis supports only a narrower context.

Representativeness

Simulation Data Representativeness is the degree to which an identified Simulation Input or Input Set sufficiently reflects the materially relevant characteristics of the reality segment it is intended to represent within a defined simulation.

Representativeness is not an intrinsic universal property of data.

It is relational.

Conceptually:

Data + Representation Target + Simulation Context → Representativeness
Assessment Representation Target

A Representation Target defines what the input is expected to represent.

The target may be: population, environment, geographic region, operating state, system configuration, behavior, physical condition, temporal condition, distribution, event class, failure state, or another materially relevant reality segment.

Without a defined Representation Target, the statement:

“The data is representative.”

is incomplete.

Representativeness Is Purpose-Relative

SIMULOS® establishes:

No dataset is representative in the abstract; it is representative only relative to a defined simulation inquiry and representation target.

The same dataset may therefore be:

Representative

for one simulation,

and:

Non-Representative

for another. Representativeness ≠ Accuracy

Data may be highly accurate and still be unrepresentative.

For example:

a perfectly measured dataset from dry-road driving may accurately describe dry-road conditions while being unsuitable as the sole representative basis for snow-road simulation.

Therefore:

Accuracy asks whether the recorded information is correct.

Representativeness asks whether the information sufficiently reflects what the simulation needs to represent. Representativeness ≠ Volume

A billion observations from one narrow operating condition may provide less

representative coverage than a much smaller dataset covering materially different relevant conditions.

SIMULOS® establishes:

Data volume cannot compensate for missing representation of materially relevant reality.

Population Representativeness

Where a simulation represents a population, SDRM determines whether materially relevant population characteristics are sufficiently represented.

Population may include: humans, vehicles, machines, organizations, infrastructure assets, biological entities, software systems, autonomous agents, devices, or other classes of objects.

Relevant dimensions may include: subgroup, capability, age, configuration, size, performance class, technical generation, behavioral profile, or another simulation-relevant distinction.

Subgroup Representation

A population-level dataset may conceal weak coverage of materially relevant subgroups.

SDRM therefore asks:

Which subgroups materially affect simulation behavior, and are they sufficiently represented?

Overall population coverage does not automatically establish subgroup coverage.

Environmental Representativeness

Environmental data should represent the environments within which simulated behavior is expected to be examined.

Relevant environmental dimensions may include: weather, terrain, temperature, lighting, infrastructure, electromagnetic conditions, network conditions, traffic density, physical obstacles, or other environmental factors.

Geographic Representativeness

Data obtained in one geographic region may not sufficiently represent another where material differences exist.

Possible differences include: infrastructure, climate, terrain, regulation, traffic behavior, architecture, population characteristics, or environmental conditions.

Geographic transfer should therefore remain explicit where material.

Operational-State Representativeness

Data from one operational state should not automatically represent another.

Relevant states may include:

Normal

Degraded

Failure

Emergency

Recovery

Maintenance

Startup

Shutdown

or another defined operating state.

A simulation intended to explore failures may require data that represents more than normal operation.

System-Version Representativeness

Simulation data may originate from a different version or configuration of the represented system.

SDRM asks whether differences between:

Data-Origin System

and:

Simulation-Represented System

are material to simulation behavior.

Behavioral Representativeness

Where behavior affects simulation outcomes, input data should sufficiently represent materially relevant behavioral variation.

This may include: human behavior, autonomous behavior, organizational behavior, market behavior, system response, interaction behavior, or adversarial behavior.

Distributional Representativeness

Representativeness includes more than averages.

Relevant characteristics may include: mean, variance, distribution shape, tails, correlations, clusters, multimodality, extremes, and conditional

distributions.

Average ≠ Representative Distribution

SIMULOS® establishes:

A representative average does not establish a representative distribution.

Two populations may have the same average while producing materially different simulation behavior.

Range Representativeness

The observed range of an input should be compared with the range relevant to the simulation.

For example:

Training / observed temperature range:
0°C–30°C

Simulation scope: –30°C–50°C

The input basis does not fully represent the simulation range.

Variation Representativeness

Simulation behavior may depend upon variation rather than absolute values alone.

SDRM therefore evaluates whether material variation patterns are sufficiently represented.

This may include: rapid changes, cyclic changes, random variation, seasonal variation, load variation, behavioral variation, or environmental variation.

Rare-Condition Representation

Rare events may have low frequency but high simulation importance.

Examples include: catastrophic failures, extreme weather, unusual human behavior, collision conditions, cascading infrastructure failures, unusual system interactions, or low-frequency safety events.

SIMULOS® establishes:

Low frequency does not imply low representational importance.

Consequence-Weighted Representativeness

The required representation of a condition may increase where failure to represent it could materially affect a high-consequence simulation conclusion.

This does not mean every rare condition must always be included.

It means representativeness decisions should reflect the purpose and consequence of the simulation.

Boundary-Condition Representativeness

Simulation behavior may change near operational boundaries.

Data concentrated in the center of an operating range may therefore fail to represent behavior near: maximum load, minimum load, speed limits, thermal limits, stability limits, communication limits, control limits, or other material boundaries.

Representation Coverage

Representation Coverage describes how much of the materially relevant Representation Target is sufficiently represented by the available input basis.

Coverage may be assessed across multiple dimensions rather than reduced to one percentage.

Multidimensional Representativeness

A dataset may be representative along one dimension and weak along another.

For example:

Population Coverage — Strong

Geographic Coverage — Strong

Failure-State Coverage — Weak

Extreme-Weather Coverage — Missing

SDRM preserves these distinctions.

No Single-Score Requirement

SDRM does not require representativeness to be compressed into one numerical score.

A single score may hide material gaps.

Representativeness may instead be represented through: dimensional assessments, coverage maps, statuses, conditions, restrictions, and gap records.

Representation Gap

A Simulation Data Representativeness Gap exists where available information does not sufficiently represent a materially relevant part of the Representation Target.

Examples include: missing population, missing environment, missing state, missing behavior, missing distribution tail, missing failure condition, missing configuration, or insufficient range.

Representation Gap Classification

A gap may be classified as:

Non-Material

The gap is not expected to materially affect the defined simulation inquiry.

Material but Bounded

The gap matters but its effect can be sufficiently restricted or isolated.

Conditional

Simulation use remains possible only under explicit conditions.

Material and Unresolved

The gap materially weakens the simulation basis and has not been sufficiently addressed.

Representativeness Undetermined

Available information is insufficient to determine materiality or coverage.

Underrepresentation

Underrepresentation occurs where a relevant condition exists in the target reality but appears insufficiently in the simulation input basis.

Underrepresentation can produce artificially weak exposure to that condition.

Overrepresentation

Overrepresentation occurs where a condition appears disproportionately in the input basis relative to the Representation Target.

This may distort simulation behavior if the simulation interprets the data as naturally distributed.

Representation Imbalance

A dataset may contain all required classes while still representing them in proportions inconsistent with the intended simulation context.

SDRM therefore distinguishes:

Presence

from:

Representative Proportion.

Intentional Oversampling

Overrepresentation is not always a defect.

Rare or high-consequence conditions may intentionally be oversampled for testing.

Where intentional, the difference between:

Natural Representation

and:

Test Amplification

should remain explicit.

Proxy Representation

A Proxy Input represents a target indirectly because direct data is unavailable or impractical.

For example:

data from one machine type may be used as a proxy for another similar machine.

Proxy use may be legitimate.

SDRM requires the representational relationship to remain explicit. Proxy ≠ Target

SIMULOS® establishes:

Similarity between a proxy and a target does not establish representational equivalence.

Material differences must remain visible.

Cross-Context Transfer

Data may be transferred from one context to another.

Conceptually:

Source Context → Transfer Basis → Target Simulation Context

SDRM governs whether the transfer preserves sufficient representativeness.

Cross-Context Representativeness Conditions

Transfer may depend upon similarities in: population, environment, system configuration, behavior, operating range, physical properties, or causal structure.

Where differences matter, restrictions should follow the transferred data.

Synthetic Representation

Synthetic data may intentionally fill representativeness gaps.

For example:

real-world datasets may contain too few critical failure events.

Synthetic data may increase coverage.

SDRM asks:

Does the synthetic information sufficiently represent the intended condition?

Its synthetic origin remains governed under SIIM. Synthetic Coverage $\neq$ Real-World Frequency

Synthetic datasets may deliberately alter frequencies.

Therefore:

Synthetic Scenario Coverage

should not automatically be interpreted as:

Observed Real-World Distribution.

Representativeness and Simulation Assumptions

Simulation Assumptions may influence representativeness requirements.

For example:

if the simulation assumes operation only within a narrow environmental range, data outside that range may not be required.

If the scope expands, the representativeness requirement may change.

Representativeness and Reality Representation

RRS — Reality Representation Standard determines which aspects of reality the simulation must represent.

SDRM determines whether the input basis sufficiently represents those selected aspects.

The relationship is:

Selected Reality → Representation Requirement → Input Representation Target →
Representativeness Assessment Representativeness and Scenario Design

SDRM does not determine which scenarios must be executed.

However, representativeness gaps may become important inputs to SSS —
Simulation Scenario Standard.

For example:

a weakly represented rare failure may justify deliberate scenario coverage.

Representativeness Materiality

Not every representational difference matters.

Materiality depends upon whether the difference could reasonably affect:
simulation behavior, simulation outputs, scenario conclusions, downstream
validation, or operational reliance.

Materiality Assessment

A representativeness gap should be evaluated relative to:

Simulation Object

Simulation Purpose

Simulation Scope

Reality Representation

Simulation Assumptions

Model & Environment

and where relevant:

intended downstream reliance.

Representativeness Restriction

An input may remain usable despite limited representativeness if its use is
explicitly restricted.

For example:

Representative only for dry-road daytime operation.

The restriction should propagate to downstream simulation interpretation.

Representativeness Condition

A Simulation Input may receive a condition such as:

Suitable only when ambient temperature remains between 0°C and 30°C.

Conditional representativeness should not become unconditional merely because simulation execution succeeds.

Representativeness Status

A material Simulation Input or Input Set may receive a status such as:

Representative

Conditionally Representative

Partially Representative

Non-Representative

or:

Representativeness Undetermined

The status is always relative to a defined Representation Target and simulation context.

No Universal Representative Status

SIMULOS® establishes:

Representative for one simulation does not mean representative for all simulations.

Representativeness status must therefore remain context-bound.

Input-Level vs Input-Set

Representativeness

Individual inputs may be representative while the complete Input Set is not.

Conversely, several individually incomplete datasets may together provide sufficient representation.

SDRM therefore supports assessment at:

Individual Input Level

and:

Simulation Input Set Level.

Complementary Representation

Multiple datasets may complement one another.

For example:

Dataset A — Normal Operation Dataset B — Failure Operation Dataset C — Extreme Weather

may collectively provide stronger representation than any dataset alone.

Representativeness Conflict

Different datasets may imply materially different representations of the same target.

For example:

Dataset A: failure frequency 0.1%

Dataset B: failure frequency 2.5%

The conflict should remain visible rather than being silently averaged away.

Representation Basis

A Simulation Representation Basis records why an input or Input Set is considered sufficiently representative for a defined simulation context.

It may reference: target definition, coverage, gaps, materiality, restrictions, conditions, proxies, synthetic supplementation, and unresolved uncertainty.

Representativeness Traceability

SDRM establishes:

Simulation Input

↓

Representation Target

↓

Relevant Representation Dimensions

↓

Coverage

↓

Gap

↓

Materiality

↓

Representativeness Status

↓

Conditions / Restrictions

↓

Simulation Use

This enables downstream users to determine what the data actually supports.

Representativeness Change

Representativeness may change even when the dataset itself does not.

For example:

the Simulation Scope may expand.

A new environment may be introduced.

A new population may be added.

The represented system may change configuration.

The same data may therefore become less representative because the target changed.

Representativeness Reassessment Trigger

Reassessment may be required when there is material change in: Simulation Object, Purpose, Scope, Reality Representation, Assumptions, Model, Environment, target population, operational conditions, system configuration, or available input basis.

No Representativeness Inheritance Principle

SIMULOS® establishes:

Representativeness must not be inherited automatically from a previous simulation when the Representation Target or simulation context has materially

changed.

No Availability-to-Representativeness Principle

Available data is not representative data by default.

No Size-to-Representativeness Principle

More data does not resolve a missing reality dimension if the additional data comes from the same unrepresentative condition.

No Majority-to-Whole Principle

Strong representation of the majority does not establish representation of every materially relevant subgroup or condition.

No Normal-to-Extreme Principle

Data representative of normal operation does not automatically represent degraded, failure, emergency, boundary, or extreme conditions.

No Proxy Equivalence Principle

Proxy data must not silently acquire the identity or representativeness status of the target it approximates.

No Successful-Execution Principle

A simulation running successfully with an input does not establish that the input was representative of the reality being simulated.

Representativeness Limitation Propagation

Material representativeness limitations should propagate downstream.

Conceptually:

Representativeness Gap → Scenario / Execution Limitation → Output
Interpretation Limitation → Downstream Reliance Limitation

The relevant downstream Parent Standards govern those later stages.

SDRM ensures the original limitation remains visible.

Minimum Implementation Framework

1. Identify the Representation Target

Define what each material Simulation Input or Input Set is expected to represent. 2. Identify Material Representation Dimensions

Determine which dimensions matter, such as: population, environment, geography, operating state, behavior, configuration, distribution, range, variation, or rare conditions.

3. Assess Representation Coverage

Determine which materially relevant portions of the target are sufficiently represented. 4. Identify Representation Gaps

Record missing, weak, imbalanced, proxy-based, or uncertain representation. 5. Assess Gap Materiality

Determine whether each gap could materially affect the defined simulation inquiry. 6. Assess Distribution and Range

Determine whether averages, ranges, tails, variation, and dependencies are sufficiently represented where material. 7. Assess Rare and Boundary Conditions

Determine whether low-frequency or boundary conditions require representation because of their simulation relevance or consequence. 8. Govern Proxy and Cross-Context Use

Document the basis and limitations where information is transferred from another population, environment, configuration, or context. 9. Assess Synthetic Supplementation

Where synthetic information fills representation gaps, preserve its role and the limits of what it represents. 10. Establish Conditions and Restrictions

Bound simulation use where representativeness is incomplete but sufficiently manageable. 11. Assign Representativeness Status

Determine whether the input basis is: Representative, Conditionally Representative, Partially Representative, Non-Representative, or Representativeness Undetermined.

12. Preserve Representativeness Traceability

Maintain the relationship between target, coverage, gaps, materiality, status, restrictions, and simulation use.

Governance Outputs

SDRM may produce: Representation Target Record, Representation Dimension Map, Population Representativeness Assessment, Subgroup Representation Record, Environmental Representativeness Assessment, Geographic Representativeness Assessment, Operational-State Representativeness Assessment, System-Version Representativeness Assessment, Behavioral Representativeness Assessment, Distributional Representativeness Assessment, Range Representativeness Record, Variation Representativeness Record, Rare-Condition Representation Record, Boundary-Condition Representation Record, Representation Coverage Map, Representation Imbalance Record, Underrepresentation Record, Overrepresentation Record, Representation Gap Record, Representation Gap Materiality Assessment, Proxy Representation Record, Cross-Context Transfer Record, Synthetic Representation Record, Complementary Representation Record, Representativeness Conflict Record, Representativeness Condition Record,

Representativeness Restriction Record, Representativeness Status, Simulation Representation Basis, Representativeness Reassessment Trigger, Representativeness Change Record, and Simulation Data Representativeness Traceability Map.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous vehicle simulation is intended to assess operation across: daytime and nighttime, dry and wet roads, urban and rural environments, normal traffic, emergency braking, and pedestrian interactions.

The available driving dataset contains millions of kilometers of data.

However, almost all observations come from:

daytime + dry roads + urban operation. Application

SDRM does not treat dataset size as proof of representativeness.

It identifies material gaps in: nighttime conditions, wet-road behavior, rural environments, emergency braking, and uncommon pedestrian interactions.

Result

The dataset receives:

Partially Representative

status for the declared simulation scope.

Its use may remain legitimate for defined urban dry-road scenarios while additional real or synthetic information is required for broader scenario coverage. Use Case 2 — Spacecraft Landing Simulation

Scenario

A spacecraft landing simulation relies upon atmospheric and terrain information derived primarily from previously observed regions.

The planned landing zone contains terrain and atmospheric conditions that are only partially represented by those observations. Application

SDRM establishes:

Observed Region ≠ Landing Target Region

and evaluates: terrain coverage, atmospheric ranges, geographic transfer, boundary conditions, and proxy assumptions.

Result

The available input basis may be classified:

Conditionally Representative

with explicit limitations requiring additional synthetic or model-derived coverage for materially underrepresented landing conditions.

Architectural Position

Simulation Data Representativeness is the second internal governance space of the Simulation Input & Data Standard (SIDS).

The complete SIDS progression is:

Simulation Input Identification

↓

Simulation Data Representativeness

↓

Simulation Data Transformation

↓

Simulation Temporal Data

↓

Simulation Input Uncertainty

SIIM determines:

What information is entering the simulation?

SDRM determines:

What reality can that information legitimately represent?

The next module, SDTM — Simulation Data Transformation Module, governs what happens when that identified and representationally understood information must be transformed before simulation use.

Foundational Principle

Simulation data becomes representative only relative to a defined representation target, and no simulation may legitimately generalize its input basis beyond the populations, environments, states, behaviors, distributions, conditions, and boundaries that the available information can sufficiently represent.

Canonical Closing Statement

Simulation Data Representativeness Module establishes the representational governance layer of the Simulation Input & Data Standard by determining whether identified Simulation Inputs and Input Sets sufficiently represent the populations, environments, geographic contexts, operating states, behaviors, configurations, distributions, ranges, variations, rare events, and boundary conditions required by a defined simulation inquiry. Through explicit Representation Targets, multidimensional coverage assessment, gap and materiality analysis, proxy governance, synthetic supplementation, conditional applicability, restrictions, reassessment triggers, and representativeness traceability, SDRM prevents data availability, volume, accuracy, technical compatibility, or successful execution from being mistaken for sufficient representation of the reality a simulation is intended to examine.

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